

Regression

What is Regression?

Process of predicting unknown value of one variable using known value of other variable

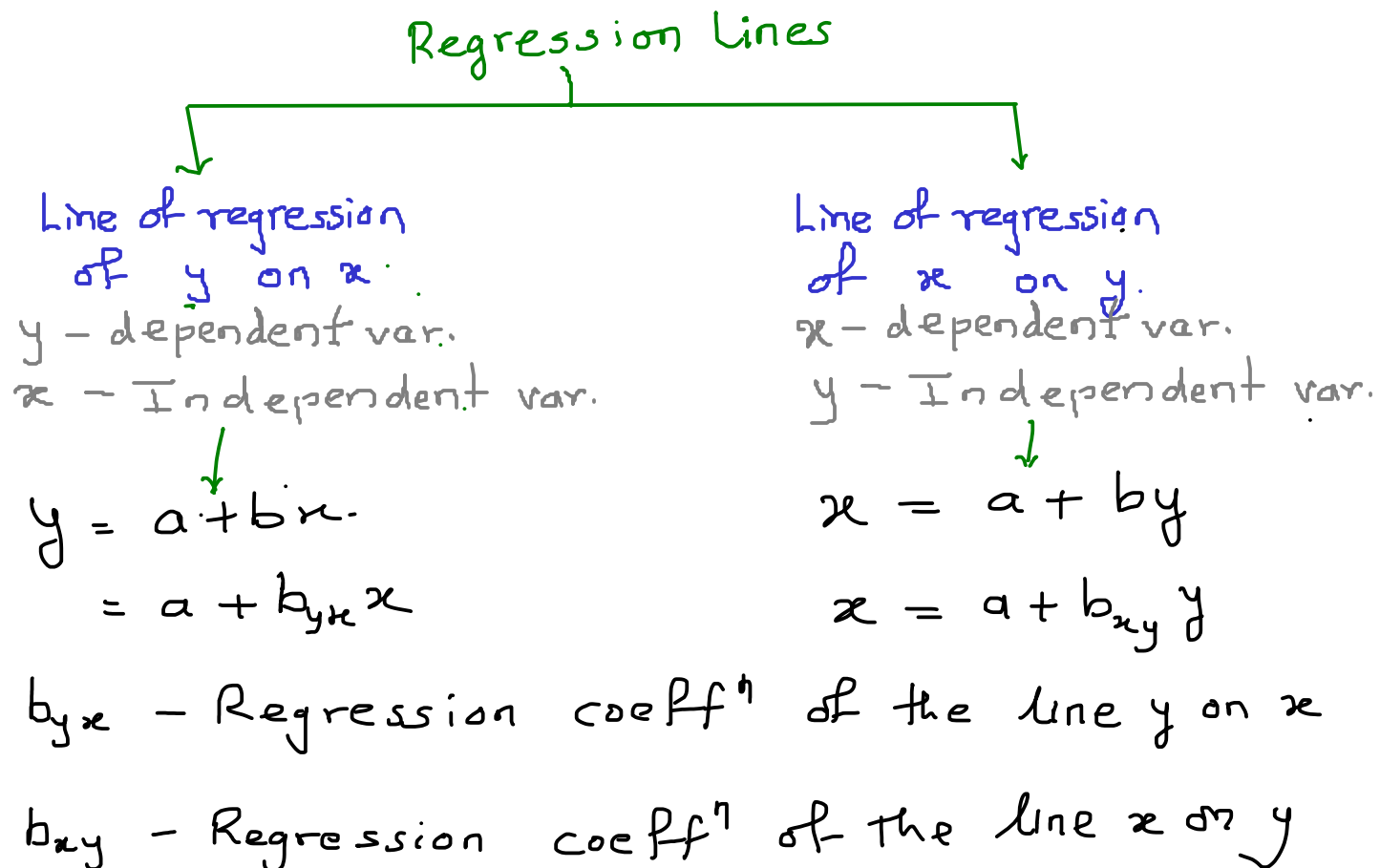
Note:-

To find the regression, two variables need to be correlated with each other

Linear Regression

If two variable are correlated then the scatter diagram will be more or less concentrated around a curve. This curve is called the Regression curve and the relationship is said to be expressed by means of curvilinear regression.

In particular, if this curve is a straight line then it is said to be Linear Regression.



I) Line of regression of y on x

Let the line of regression of y on x is given by

$$y = a + bx \quad \text{--- (1)}$$

$a = \text{constant}$

Normal eqⁿs of eqⁿ (1) are

$$\sum_{i=1}^n a = a \sum_{i=1}^n x_i^0$$

$$\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i \quad \text{--- (2)}$$

$$= a[x_1^0 + x_2^0 + \dots + x_n^0]$$

$$= a[1 + 1 + \dots + 1] \text{ } n \text{ times}$$

$$= a \cdot n$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{--- (3)}$$

Dividing eqⁿ (2) by n , we get

$$\frac{\sum y}{n} = a + b \cdot \frac{\sum x}{n}$$

$$\bar{x} = \frac{\sum x}{n} = \text{mean of } x$$

$$\bar{y} = \frac{\sum y}{n} = \text{mean of } y$$

$$\Rightarrow \bar{y} = a + b\bar{x} \quad \text{--- (4)}$$

* Eqⁿ (4) shows that the means of x & y i.e. $\bar{x}$ & $\bar{y}$ satisfies the eqⁿ of line of regression.

Now subtracting eqⁿs (1) & (4) we get

$$y - \bar{y} = b(x - \bar{x}) = b_{yx}(x - \bar{x})$$

which represents eqⁿ of regression line of y on x .

b is called regression coefficient & it is denoted by b_{yx}

Calculation of b_{yx}

$$\sum y = na + b \sum x \quad \times \sum x$$

$$\sum xy = a \sum x + b \sum x^2 \quad \times n$$

$$n \sum xy = na \sum x + nb \sum x^2$$

$$- \sum x \sum y = -na \sum x + b (\sum x)^2$$

$$n \sum xy - \sum x \sum y = b [n \sum x^2 - (\sum x)^2]$$

Divide both sides by n^2

$$\frac{n \sum xy - \sum x \sum y}{n^2} = b \left[\frac{n \sum x^2}{n^2} - \frac{(\sum x)^2}{n^2} \right]$$

$$\frac{\sum xy}{n} - \frac{\sum x}{n} \cdot \frac{\sum y}{n} = b \left[\frac{1}{n} \sum x^2 - \left(\frac{\sum x}{n} \right)^2 \right]$$

$$\frac{1}{n} \sum xy - \bar{x} \bar{y} = b \left[\frac{1}{n} \sum x^2 - \bar{x}^2 \right]$$

$$b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum x^2 - \bar{x}^2}$$

$$b_{yx} = \frac{\text{cov}(x, y)}{\text{Var.}(x)} = \frac{\text{cov.}(x, y)}{\sigma_x^2}$$

cov - co-variance

Var - Variance.

Similarly

$$b_{xy} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum y^2 - \bar{y}^2}$$

$$b_{xy} = \frac{\text{cov}(x, y)}{\text{var.}(y)} = \frac{\text{cov}(x, y)}{\sigma_y^2}$$

Consider the product of regression coeffs

$$b_{xy} b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\left[\frac{1}{n} \sum x^2 - \bar{x}^2 \right]} \times \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\left[\frac{1}{n} \sum y^2 - \bar{y}^2 \right]}$$

$$= \frac{\left[\frac{1}{n} \sum xy - \bar{x} \bar{y} \right]^2}{\left[\sqrt{\frac{1}{n} \sum x^2 - \bar{x}^2} \sqrt{\frac{1}{n} \sum y^2 - \bar{y}^2} \right]^2}$$

$$= \left[\frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\sqrt{\frac{1}{n} \sum x^2 - \bar{x}^2} \sqrt{\frac{1}{n} \sum y^2 - \bar{y}^2}} \right]^2$$

$$\Rightarrow b_{xy} b_{yx} = r^2$$

$$\Rightarrow r = \sqrt{b_{yx} b_{xy}}$$

Geometric mean

Properties of regression coefficients

- 1) Correlation coeffⁿ is geometric mean between the regression coeffⁿs.

i.e.
$$r = \sqrt{b_{yx} \cdot b_{xy}}$$

- 2) If one of the regression coeffⁿ is greater than 1 then other must be less than 1
i.e. if $b_{yx} > 1$ then $b_{xy} < 1$

or if $b_{xy} > 1$ then $b_{yx} < 1$

- 3) Arithmetic mean of regression coeffⁿs is greater than correlation coeffⁿ.

i.e.
$$\frac{b_{yx} + b_{xy}}{2} > r$$

- 4) Regression coeffⁿs are independent of the origin but not of scale.

- 5) The correlation coeffⁿ & the two regression coeffⁿs have same sign.

Regression line of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where

$$b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum x^2 - \bar{x}^2}$$
$$= \frac{\text{cov}(x, y)}{\text{Var}(x)} = \frac{\text{cov}(x, y)}{\sigma_x^2}$$

Regression line of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$b_{xy} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum y^2 - \bar{y}^2}$$
$$= \frac{\text{cov}(x, y)}{\text{Var}(y)} = \frac{\text{cov}(x, y)}{\sigma_y^2}$$

Var $\rightarrow$ Variance.

Variance = (standard deviation)²

- ① $r = \pm \sqrt{b_{yx} b_{xy}}$
- ② Both b_{yx} & b_{xy} have same sign
- ③ Sign of r is same as sign of b_{yx} & b_{xy}
- ④ If $b_{yx} > 1$ then $b_{xy} < 1$ Or vice versa
- ⑤ $\bar{x}, \bar{y}$ satisfies the eqⁿs of regression lines

Examples.

1) Find the lines of regression for the following data.

x	10	14	19	26	30	34	39
y	12	16	18	26	29	35	38

and estimate y for $x = 14.5$
and x for $y = 29.5$

Solⁿ - To find

- ① $y - \bar{y} = b_{yx}(x - \bar{x})$
- ② y at $x = 14.5$
- ③ $x - \bar{x} = b_{xy}(y - \bar{y})$
- ④ x at $y = 29.5$

where

$$b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum x^2 - \bar{x}^2}$$

$$b_{xy} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum y^2 - \bar{y}^2}$$

$\therefore$ first we have to calculate

$$\sum x, \sum y, \bar{x}, \bar{y}, \sum x^2, \sum y^2, \sum xy$$

	x	y	x^2	y^2	$x \cdot y$
	10	12	100	144	120
	14	16	196	256	224
	19	18	361	324	342
	26	26	676	676	676
	30	29	900	841	870
	34	35	1156	1225	1190
	39	38	1521	1444	1482
Total	172	174	4910	4910	4904

$$\therefore \bar{x} = \frac{\sum x}{n} = \frac{172}{7} = 24.57$$

$$\bar{y} = \frac{\sum y}{n} = \frac{174}{7} = 24.85$$

① Regression line of y on x .
It is given by

$$y - \bar{y} = b_{yx} (x - \bar{x}) \quad \text{--- ①}$$

where

$$b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum x^2 - \bar{x}^2} = \frac{\frac{4904}{7} - \frac{172}{7} \times \frac{174}{7}}{\frac{4910}{7} - \left(\frac{172}{7}\right)^2}$$

$$= 0.9193$$

Putting this value in eqn ① we get

$$y - \frac{174}{7} = 0.9193 \left(x - \frac{172}{7} \right)$$

$$\Rightarrow y = 0.9193x + 2.2686$$

Now when $x = 14.5$

$$y = 0.9193(14.5) + 2.2686$$

$$y = 15.5984$$

⑪ Regression line of x on y
It is given by

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad \text{--- (2)}$$

where

$$b_{xy} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum y^2 - \bar{y}^2} = \frac{\frac{4904}{7} - \frac{172}{7} \times \frac{174}{7}}{\frac{4910}{7} - \left(\frac{174}{7}\right)^2}$$
$$= 1.0747$$

$\therefore$ eqn (2) becomes

$$x - \frac{172}{7} = 1.0747 \left(y - \frac{174}{7} \right)$$

$$\Rightarrow x = 1.0747 y - 2.1425$$

when $y = 29.5$

$$x = 1.0747(29.5) - 2.1425$$

$$x = 29.5611$$

- 2) Determine the eqⁿs of regression lines for the following data. Also find the value of
 (i) y for $x = 4.5$ (ii) x when $y = 13$

x	2	3	5	7	9	10	12	15
y	2	5	8	10	12	14	15	16

	x	y	x^2	y^2	$x \cdot y$
	2	2	4	4	4
	3	5	9	25	15
	5	8	25	64	40
	7	10	49	100	70
	9	12	81	144	108
	10	14	100	196	140
	12	15	144	225	180
	15	16	225	256	240
Total	63	82	637	1014	797

Here $n = 8$

$$\therefore \bar{x} = \frac{\sum x}{n} = \frac{63}{8} = 7.875$$

$$\bar{y} = \frac{\sum y}{n} = \frac{82}{8} = 10.25$$

(i) Regression line of y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$b_{yx} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum x^2 - \bar{x}^2} = \frac{\frac{797}{8} - (7.875 \times 10.25)}{\frac{637}{8} - (7.875)^2}$$

$$= 1.0736$$

$$\therefore y - 10.25 = 1.0736(x - 7.875)$$

$$y = 1.0736x + 1.7954$$

Now when $x = 4.5$

$$y = 1.0736(4.5) + 1.7954$$

$$y = 6.6266$$

② Regression line of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where

$$b_{xy} = \frac{\frac{1}{n} \sum xy - \bar{x} \bar{y}}{\frac{1}{n} \sum y^2 - \bar{y}^2} = \frac{\frac{797}{8} - (7.875 \times 10.25)}{\frac{1014}{8} - (10.25)^2}$$

$$= 0.8718$$

$$\therefore x - 7.875 = 0.8718(y - 10.25)$$

$$\Rightarrow x = 0.8718y - 1.06095$$

when $y = 13$

$$x = 0.8718(13) - 1.06095$$

$$x = 10.2725$$

3) find the line of regression of y on x for the following data.

H/W

x	40	44	28	30	44	38	31
y	32	35	26	30	38	34	28

4) Determine the eqⁿs of regression lines for the following data. Also find the value of
 (i) y for $x = 10$ (ii) x when $y = 12.5$

H/W

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

	x	y	x^2	y^2	$x \cdot y$
	1	9	1	81	9
	2	8	4	64	16
	3	10	9	100	30
	4	12	16	144	48
	5	11	25	121	55
	6	13	36	169	78
	7	14	49	196	98
	8	16	64	256	128
	9	15	81	225	135
Total	45	108	285	1356	597

5) In a partially destroyed laboratory record of an analysis of a correlation data, following results only are eligible

Variance of $x = 9$

Regression eq's : $8x - 10y + 66 = 0$

$$40x - 18y = 244$$

Find (i) The mean values of x and y

(ii) Standard deviation of y

(iii) The correlation coeffⁿ

6) If the two lines of regression are $9x + y - \lambda = 0$
& $4x + y = u$ the means of x & y are 2 & -3
resp., Find the values of λ , u & coeffⁿ of
correlation.